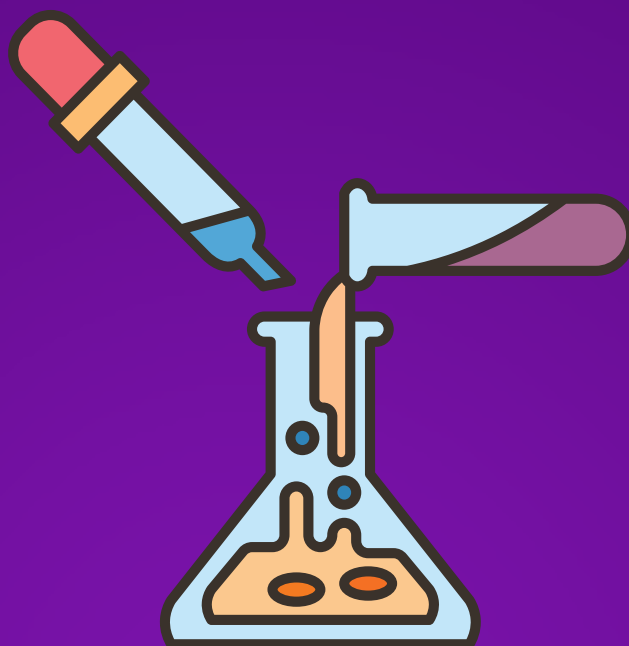




INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

f – Block Element

English Medium

10. The reason for the stability of Gd^{3+} ion is
(A) $4f$ subshell — half filled
(B) $4f$ subshell — completely filled
(C) Possesses the general electronic configuration of noble gases
(D) $4f$ subshell empty
11. Which of the following pairs has the same size ?
(A) Zn^{2+} , Hf^{4+} (B) Fe^{2+} , Ni^{2+} (C) Zr^{4+} , Ti^{4+} (D) Zr^{4+} , Hf^{4+}
12. Which of the following ions will exhibit colour in aqueous solutions ?
(A) Sc^{3+} ($Z = 21$) (B) La^{3+} ($Z = 57$) (C) Ti^{3+} ($Z = 22$) (D) Lu^{3+} ($Z = 71$)
13. Which of the following exhibits only +3 oxidation state ?
(A) Ac (B) Pa (C) U (D) Th

FB0010

FB0011

FB0012

FB0013

EXERCISE # JEE-MAIN

1. Arrange Ce^{3+} , La^{3+} , Pm^{3+} and Yb^{3+} in increasing order of their ionic radius - [AIEEE-2002]
 (1) $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{Ce}^{3+} < \text{La}^{3+}$ (2) $\text{Ce}^{3+} > \text{Yb}^{3+} < \text{Pm}^{3+} < \text{La}^{3+}$
 (3) $\text{Yb}^{3+} > \text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+}$ (4) $\text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+} > \text{Yb}^{3+}$
FB0014
2. Most common oxidation states shown by cerium are : [AIEEE-2002]
 (1) +2, +4 (2) +3, +4 (3) +3, +5 (4) +2, +3
FB0015
3. A reduction in atomic size with increase in atomic number is a characteristic of elements of :
 (1) f-Block (2) Radioactive series [AIEEE-2003]
 (3) High atomic masses (4) d-Block
FB0016
4. The radius of La^{3+} is $1.06\text{\AA}$, which of the following given values will be closest to the radius of Lu^{3+} (At no. of Lu = 71, La = 57)- [AIEEE-2003]
 (1) $1.6\text{\AA}$ (2) $1.4\text{\AA}$ (3) $1.06\text{\AA}$ (4) $0.85\text{\AA}$
FB0017
5. Cerium ($Z=58$) is an important member of the lanthanoids. Which of the following statements about cerium is **INCORRECT** - [AIEEE-2004]
 (1) Cerium (IV) acts as an oxidising agent
 (2) The +3 oxidation state of cerium is more stable than the +4 oxidation state
 (3) The +4 oxidation state of cerium is not known in solutions
 (4) The common oxidation states of cerium are +3 and +4
FB0018
6. The lanthanoid contraction is responsible for the fact that - [AIEEE-2005]
 (1) Zr and Y have about the same radius (2) Zr and Nb have similar oxidation state
 (3) Zr and Hf have about the same radius (4) Zr and Zn have similar oxidation state
FB0019
7. Lanthanoid contraction is caused due to [AIEEE-2006]
 (1) the same effective nuclear charge from Ce to Lu
 (2) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
 (3) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 (4) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge
FB0020
8. Identify the **INCORRECT** statement among the following- [AIEEE-2007]
 (1) d-block elements show irregular and erratic chemical properties among themselves
 (2) La and Lu have partially filled d-orbitals and no other partially filled orbitals
 (3) The chemistry of various lanthanoids is very similar
 (4) 4f and 5f-orbitals are equally shielded
FB0021

9. The actinoids exhibits more number of oxidation states in general than the lanthanoids. This is because
- [AIEEE-2007]

- (1) The $5f$ -orbitals are more buried than the $4f$ -orbitals
- (2) There is a similarity between $4f$ - and $5f$ in their angular part of the wave function
- (3) The actinoids are more reactive than the lanthanoids
- (4) The $5f$ -orbitals extend further from the nucleus than the $4f$ -orbitals

FB0022

10. Larger number of oxidation states are exhibited by the actinoids than those by the lanthanoids, the main reason being [AIEEE-2008]

- (1) $4f$ orbitals more diffused than the $5f$ orbitals
- (2) lesser energy difference between $5f$ and $6d$ than between $4f$ and $5d$ orbitals
- (3) more energy difference between $5f$ and $6d$ than between $4f$ and $5d$ orbitals
- (4) more reactive nature of the actinides than the lanthanides

FB0023

11. Knowing that the chemistry of lanthanoids (Ln) is dominated by its +3 oxidation state, which of the following statements is **INCORRECT** ? [AIEEE-2009]

- (1) Ln(III) compounds are generally colourless
- (2) Ln(III) hydroxides are mainly basic in character
- (3) Because of the large size of the Ln(III) ions the bonding in its compounds is predominantly ionic in character
- (4) The ionic sizes of Ln(III) decrease in general with increasing atomic number

FB0024

12. In context of the lanthanoids, which of the following statements is **NOT CORRECT** ? [AIEEE-2011]

- (1) Because of similar properties the separation of lanthanoids is not easy
- (2) Availability of $4f$ electrons results in the formation of compounds in +4 state for all the members of the series
- (3) There is a gradual decrease in the radii of the members with increasing atomic number in the series
- (4) All the members exhibit +3 oxidation state

FB0025

13. Which of the following forms stable +4 oxidation state ? [Jee-Main 2012, Online]

- (1) La($Z = 57$)
- (2) Eu($Z = 63$)
- (3) Gd($Z = 64$)
- (4) Ce($Z = 58$)

FB0026

14. The number of unpaired electrons in Gadolinium [$Z = 64$] is :- [Jee-Main 2012, Online]

- (1) 2
- (2) 6
- (3) 8
- (4) 3

FB0027

15. The lanthanide ion that would show colour is- [Jee-Main 2019, Online]
 (1) Sm^{3+} (2) La^{3+} (3) Lu^{3+} (4) Gd^{3+}
 FB0028
16. The highest possible oxidation states of uranium and plutonium, respectively, are :- [Jee-Main 2019, Online]
 (1) 6 and 4 (2) 7 and 6
 (3) 4 and 6 (4) 6 and 7
 FB0029
17. The correct order of atomic radii is : [Jee-Main 2019, Online]
 (1) $\text{Ce} > \text{Eu} > \text{Ho} > \text{N}$ (2) $\text{N} > \text{Ce} > \text{Eu} > \text{Ho}$
 (3) $\text{Eu} > \text{Ce} > \text{Ho} > \text{N}$ (4) $\text{Ho} > \text{N} > \text{Eu} > \text{Ce}$
 FB0030
18. The maximum number of possible oxidation states of actinoids are shown by [Jee-Main 2020, Online]
 (1) berkelium (Bk) and californium (Cf) (2) nobelium (No) and lawrencium (Lr)
 (3) actinium (Ac) and thorium (Th) (4) neptunium (Np) and plutonium (Pu)
 FB0031
19. The electronic configurations of bivalent europium and trivalent cerium are (atomic number : $\text{Xe} = 54$, $\text{Ce} = 58$, $\text{Eu} = 63$) [Jee-Main 2020, Online]
 (1) $[\text{Xe}] 4f^4$ and $[\text{Xe}] 4f^9$ (2) $[\text{Xe}] 4f^7$ and $[\text{Xe}] 4f^1$
 (3) $[\text{Xe}] 4f^7 6s^2$ and $[\text{Xe}] 4f^2 6s^2$ (4) $[\text{Xe}] 4f^2$ and $[\text{Xe}] 4f^7$
 FB0032
20. Which one of the following lanthanoids does not form MO_2 ? [M is lanthanoid metal]
 (1) Pr (2) Dy (3) Nd (4) Yb [J-Main 2021]
 FB0034
21. Given below are two statement : one is labelled as **Assertion A** and the other is labelled as **Reason R** :
Assertion A : Size of Bk^{3+} ion is less than Np^{3+} ion. [J-Main 2021]
Reason R : The above is a consequence of the lanthanoid contraction.
 In the light of the above statements, choose the correct answer from the options given below :
 (1) **A** is false but **R** is true
 (2) Both **A** and **R** are true but **R** is not the correct explanation of **A**
 (3) Both **A** and **R** are true and **R** is the correct explanation of **A**
 (4) **A** is true but **R** is false
 FB0035

22. Which one of the lanthanoids given below is the most stable in divalent form? [J-Main 2022]
 (1) Ce (Atomic Number 58)
 (2) Sm (Atomic Number 62)
 (3) Eu (Atomic Number 63)
 (4) Yb (Atomic Number 70) **FB0038**
23. The most common oxidation state of Lanthanoid elements is +3. Which of the following is likely to deviate easily from +3 oxidation state? [J-Main 2022]
 (1) Ce (At. No. 58)
 (2) La (At. No. 57)
 (3) Lu (At. No. 71)
 (4) Gd (At. No. 64) **FB0039**

ANSWERS KEY

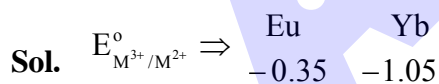
EXERCISE # O-1

1	2	3	4	5	6	7	8	9	10
D	B	B	C	D	B	B	A	B	A
11	12	13							
D	C	A							

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	1	4	3	3	2	4	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	4	3	1	4	3	4	2	4
Que.	21									
Ans.	4									

22. (3)



Hence, due to more reduction potential in Eu as compared to Yb, it can be concluded that Eu^{2+} is more stable than Yb^{2+} .

23. (1)

